BIOMARKERX CLINICAL SEPSIS EARLY WARNING REPORT

Dual-Tier Asymmetric AI Risk Engine (v1.0.0_canonical_lock)

Emory University Hospital / General ICU
Attending: Dr. Sarah Jenkins, MD
Date: 2026-10-04 02:33:37

Patient Name: John Doe	**MRN / Patient ID:** MRN-984210	**Age / Gender:** 64 yrs / Male	**ICU Location:** Medical ICU - Bed 12

CALIBRATED SEPSIS RISK PROBABILITY
38.5%
Category: CRITICAL HIGH RISK

DUAL-TIER POLICY STATE OUTPUT (VAL_POL_0450)
- **Tier 1 Screening Tier (τ_s=0.015): ACTIVE SURVEILLANCE ALERT**
- **MAP Trend Gate (MAP $\leq$ 75 $\vee$ Δ_3MAP $\leq$ -3.0): PASSED (GATED)**
- **Tier 2 Bedside Confirmation Alarm (τ_c=0.300, p=3): BEDSIDE AUDIBLE ALARM TRIGGERED**

IN-DEPTH CLINICAL DIAGNOSTIC SYNTHESIS & PATHOPHYSIOLOGICAL RATIONALE

Diagnostic Synthesis (Time t=0h): Patient present with a calibrated sepsis risk probability of **38.5%** (CRITICAL HIGH RISK). Pathophysiologically, this elevated risk state is primarily driven by acute microvascular hypoperfusion and an uncompensated Systemic Inflammatory Response Syndrome (SIRS). Specifically, the patient demonstrates marked compensatory tachycardia (Heart Rate: 108.0 bpm, Z-score velocity > +0.3/hr) combined with acute arterial hypotension (MAP: 64.0 mmHg, with a 3-hour backward trajectory drop of -4.5 mmHg), yielding an elevated Shock Index of 1.17 (population median: 0.72). Concurrently, serum blood lactate is elevated at 2.8 mmol/L (normal range: 0.5 - 1.6 mmol/L), signaling cellular dysoxia and anaerobic glycolytic metabolic shift secondary to systemic vasodilation and tissue hypoperfusion. Immunologically, leukocytosis (WBC: 16.4 × 10^3/μL) and tachypnea (Respiratory Rate: 26.0 breaths/min) reflect active neutrophil demargination and respiratory compensation for metabolic acidosis. Under the continuous temporal trajectory engine (v1.0.0_canonical_lock), the combination of trajectory Z-score velocity (dZ/dt < -0.4/hr for MAP) and 3-consecutive-window confirmation persistence confirms early acute organ dysfunction targeted within the 4-hour forward lookahead window (S_{next4}).

12-HOUR TEMPORAL RISK TRAJECTORY & FORWARD TARGET WINDOW (S_{next4})

t=-12h	t=-9h	t=-6h	t=-3h	t=0h	t+1h	t+2h	t+3h	t+4h
20.5%	25.0%	29.5%	34.0%	38.5%	41.6%	44.9%	48.5%	52.4%
Historical	Historical	Historical	Historical	Historical	Target S4	Target S4	Target S4	Target S4

PRIMARY RISK DRIVERS (+SHAP) VS PROTECTIVE FACTORS (-SHAP)

Factor & Observed Value	Population Median & Norm	TreeSHAP Weight	Clinical Risk Direction
Systolic BP: 92.0 mmHg	Median: 118.5 mmHg Norm: 90 - 140	**+2.12 SHAP**	↑ **INCREASES RISK**
Heart Rate: 108.0 bpm	Median: 84.0 bpm Norm: 60 - 90	**+1.92 SHAP**	↑ **INCREASES RISK**
Mean Arterial Pressure: 64.0 mmHg	Median: 77.0 mmHg Norm: 70 - 105	**+1.26 SHAP**	↑ **INCREASES RISK**
Respiratory Rate: 26.0 breaths/min	Median: 18.0 breaths/min Norm: 12 - 20	**+0.64 SHAP**	↑ **INCREASES RISK**
Blood Lactate: 2.8 mmol/L	Median: 1.7 mmol/L Norm: 0.5 - 1.6	**+0.18 SHAP**	↑ **INCREASES RISK**

Protective Factors	N/A	0.000 SHAP	No significant protective drivers detected.

122 FEATURE MANIFEST & CONTINUOUS STREAM TESTING SPECIFICATION

• **122-Feature Manifest Scope:** The model evaluates 122 verified features spanning 39 raw vitals/labs (HR, SBP, MAP, Temp, Resp, WBC, Lactate, Creatinine, BUN, Platelets, AST, Bilirubin, Glucose, PaCO2, pH, BaseExcess, etc.), 39 patient-relative Z-scores (z_{i,t}), 39 causal backward deltas (dZ/dt), and 5 derived clinical indices (Shock Index, BUN/Creatinine ratio, Oxygenation Index).
• **Continuous Streaming Protocol:** Evaluated across hourly sliding windows (t=1..T) with expanding temporal history (x_{<=t}) and zero future lookahead leak.
• **Cross-Institution Validation:** Validated on Emory Set A (N=2,974) and holdout BIDMC Set B (N=20,000) with 0.00% patient overlap, achieving 30.52% transportability gap reduction.
• **Research Disclaimer:** Retrospective research AI system (v1.0.0_canonical_lock). This report is for scientific verification and does NOT replace physician judgment.